

ICT703 Programming

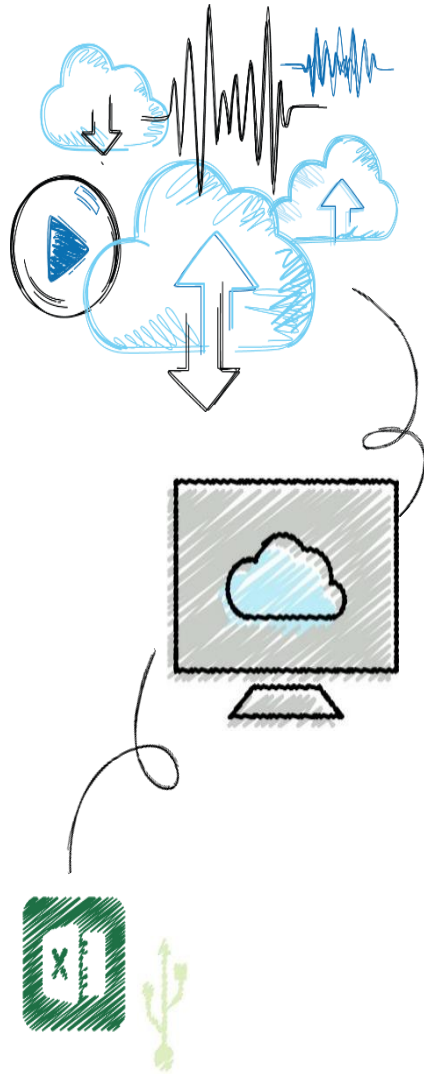
Module 8

File Processing



Objectives

- Review the basic file operation functions
- Investigate more robust file processing with csv files.
- Use modules to processing more file types





Text Files

- A text file is software object that stores data on permanent medium
 - Basic Operations are:
 - Open a file (can be to read, write or append)
 - **f = open("myfile.txt", 'r')**
 - Process a file
 - **f.readline()**
 - Close the file
 - **f.close()**
-



More than Basic Files

- Typically, we want to process the file with some structure.
- What about a csv file or an Excel file
- What about reading writing a dictionary data structure
- What about extracting data from a more structured source like a database?



Reading a CSV File – Plan A

- Assume we have an input file that looks like the following
App, Category, Rating, Reviews, Size
Photo Editor, ART_AND_DESIGN, 4.1, 159, 19M
- We could open the file, and work on every line (instead of printing it)

```
infile = open("app_file.csv", "r")  
for line in infile:  
    print(line)
```

Problem is we must find the commas, and decompose the line into the components



Reading a CSV File – Plan B

- So why not use a string to list method and then we will have the components

- Open the file, split the lines by commas and then process it

```
infile = open("app_file.csv", "r")
```

```
for line in infile:
```

```
    line_list = line.strip().split(",")
```

```
    print(line_list)
```

- Problem is the case where the data has a , (within a quote)

App, Category, Rating, Reviews, Size

"Photo, Video, Picture Editor", ART_AND_DESIGN, 4.1, 159, 19M

This will split the line into too many components



Reading a CSV File – Plan C

- There is a module for that !
 - Open the file, split the lines by commas and then process it

```
import csv
infile = open("app_file.csv", "r")
csv_reader = csv.reader(infile)
for line_list in csv_reader:
    print(line_list)
```
 - This looks better – now the line is split into the component parts (in a list) just like we would expect from a csv file
 - Problem is we have to access each component by the column number
 - Eg line_list[4] is the review component
 - Also we need to ignore the header row
-



Reading a CSV File – Plan D

- Let' try another method from the csv module
- Open the file, split the lines by commas and then process it

```
import csv
infile = open("app_file.csv", "r")
csv_dict_reader = csv.DictReader(infile)
for line_dict in csv_dict_reader:
    print(line_dict)
```

- This looks even better – now the line is split into the component parts (in a dictionary) so we can access any column by its column name. The first line becomes the key for the dictionary
 - `line_dict["Size"]` has the value "19M"
-



Reading an Excel File

- We can go further and read an Excel file – which is perfect for structured data.
- We will use the Pandas module which is extremely powerful and we will revisit again.
- Open the file, split the lines by commas and then process it

```
import pandas as pd
df = pd.read_excel('Google Play Store.xlsx')
print(df)
```

- This returns the whole spreadsheet as a dataframe and we can access elements quite easily. This gives the first row in the category column.

```
print(df['Category'][1])
```

- More on Pandas Later
-



Writing/Reading a Dictionary

- A dictionary data structure is perfect for configuration information within a program. For example, player information in a game:

```
player1 = {'name': 'Andrew', 'high score': 350, 'current level': 3}
```

- We want to save that in a file at the end of the game as a dictionary.

```
import json  
player1 = {'name': 'Andrew', 'high score': 350, 'current level': 3}  
json.dump(player1, open("player_config.txt", 'w'))
```

- When we start the game, we can load up the configuration directly into the dictionary.

```
player1 = json.load(open("player_config.txt"))  
print (player1)
```



Writing/Reading to a database

- There are many database platforms (relational like SQL, or NoSQL databases like MongoDB) that are perfect for storing our data. Python has some great modules to access these with easy.
- Stay Tuned !